

EACKNATH SIR'S

50 PRACTICE QUESTIONS

ON NUMBER SYSTEM

Student Practice Worksheet

1. Check Whether a Number is Even or Odd

Problem Statement

Given an integer N, determine whether the given number is even or odd. A number is even if it is completely divisible by 2. Otherwise, it is odd.

Input

A single integer N.

Output

Print Even if the number is even; otherwise, print Odd.

Examples

Input:

N = 24

Output:

Even

Explanation:

$24 \% 2 = 0$, therefore 24 is an Even Number.

Input:

N = 17

Output:

Odd

Explanation:

$17 \% 2 = 1$, therefore 17 is an Odd Number.

2. Check Whether a Number is Positive, Negative, or Zero

Problem Statement

Given an integer N, determine whether it is positive, negative, or zero.

Input

A single integer N.

Output

Print Positive, Negative, or Zero.

Examples

Input:

N = 15

Output:

Positive

Explanation:

15 is greater than zero.

Input:

N = -8

Output:

Negative

Explanation:

-8 is less than zero.

Input:

N = 0

Output:

Zero

Explanation:

The number is equal to zero.

3. Find the Greatest of Two Numbers

Problem Statement

Given two integers, find the greater number.

Input

Two integers A and B.

Output

Print the greater number. If both are equal, print the common value.

Examples

Input:

A = 25, B = 18

Output:

25

Explanation:

25 is greater than 18.

4. Find the Greatest of Three Numbers

Problem Statement

Given three integers, find the greatest among them.

Input

Three integers A, B, and C.

Output

Print the greatest number.

Examples

Input:

A = 25, B = 48, C = 31

Output:

48

Explanation:

48 is greater than both 25 and 31.

5. Check Whether a Year is a Leap Year

Problem Statement

Given a year, determine whether it is a leap year. A year is a leap year if it is divisible by 400, or if it is divisible by 4 but not divisible by 100.

Input

A single integer representing the year.

Output

Print Leap Year or Not a Leap Year.

Examples

Input:

Year = 2000

Output:

Leap Year

Explanation:

2000 is divisible by 400.

Input:

Year = 1900

Output:

Not a Leap Year

Explanation:

1900 is divisible by 100 but not by 400.

6. Find the Sum of First N Natural Numbers

Problem Statement

Given a positive integer N, find the sum of the first N natural numbers.

Input

A single integer N.

Output

Print the sum.

Examples

Input:

N = 10

Output:

55

Explanation:

$1 + 2 + 3 + \dots + 10 = 55.$

7. Count the Number of Digits in an Integer

Problem Statement

Given an integer N, count how many digits it contains.

Input

A single integer N.

Output

Print the number of digits. Note: 0 contains one digit.

Examples

Input:

N = 58392

Output:

5

Explanation:

58392 contains five digits.

8. Find the Sum of Digits of a Number

Problem Statement

Given an integer N, find the sum of all its digits.

Input

A single integer N.

Output

Print the sum of its digits.

Examples

Input:

N = 529

Output:

16

Explanation:

$5 + 2 + 9 = 16$.

Input:

N = 101

Output:

2

Explanation:

$1 + 0 + 1 = 2$.

9. Reverse the Digits of a Number

Problem Statement

Given an integer N, reverse its digits.

Input

A single integer N.

Output

Print the reversed number.

Examples

Input:

N = 12345

Output:

54321

Explanation:

The digits of 12345 in reverse order are 54321.

Input:

N = 10400

Output:

401

Explanation:

Leading zeros are not represented in an integer.

10. Check Whether a Number is a Palindrome

Problem Statement

Given an integer N, determine whether it reads the same forward and backward.

Input

A single integer N.

Output

Print Palindrome or Not Palindrome.

Examples

Input:

N = 121

Output:

Palindrome

Explanation:

121 remains the same when reversed.

Input:

N = 123

Output:

Not Palindrome

Explanation:

Reverse of 123 is 321, which is different.

11. Find the Maximum and Minimum Digit in a Number

Problem Statement

Given an integer N, find the largest and smallest digit present in the number.

Input

A single integer N.

Output

Print the maximum and minimum digit.

Examples

Input:

N = 583920

Output:

Maximum Digit = 9, Minimum Digit = 0

Explanation:

The largest digit is 9 and the smallest digit is 0.

12. Count the Occurrences of a Given Digit

Problem Statement

Given an integer N and a digit D, count how many times digit D occurs in N.

Input

An integer N and a digit D.

Output

Print the number of occurrences of D.

Examples

Input:

N = 1223332, D = 3

Output:

3

Explanation:

The digit 3 occurs three times.

13. Find the Frequency of Every Digit

Problem Statement

Given an integer N, find how many times each digit from 0 to 9 occurs.

Input

A single integer N.

Output

Print the frequency of each digit from 0 to 9.

Examples

Input:

N = 112233

Output:

0 → 0, 1 → 2, 2 → 2, 3 → 2, and all other digits → 0

Explanation:

Digits 1, 2, and 3 each occur twice.

14. Replace All 0s with 1s in a Number

Problem Statement

Given an integer N, replace every 0 digit with 1.

Input

A single integer N.

Output

Print the modified number.

Examples

Input:

N = 102003

Output:

112113

Explanation:

Every 0 is replaced by 1: 1 0 2 0 0 3 → 1 1 2 1 1 3.

Input:

N = 204

Output:

214

Explanation:

The 0 is replaced by 1.

15. Remove a Given Digit from a Number

Problem Statement

Given an integer N and a digit D, remove all occurrences of D from the number.

Input

An integer N and a digit D.

Output

Print the resulting number.

Examples

Input:

N = 122342, D = 2

Output:

134

Explanation:

All occurrences of digit 2 are removed.

16. Find the Product of Digits of a Number

Problem Statement

Given an integer N, find the product of all its digits.

Input

A single integer N.

Output

Print the product of its digits.

Examples

Input:

N = 234

Output:

24

Explanation:

$2 \times 3 \times 4 = 24$.

17. Repeatedly Add Digits Until a Single Digit Remains

Problem Statement

Given an integer N, repeatedly add all its digits until the result contains only one digit.

Input

A single integer N.

Output

Print the final single digit.

Examples

Input:

N = 529

Output:

7

Explanation:

$5 + 2 + 9 = 16$, then $1 + 6 = 7$. Since 7 is a single digit, return 7.

Input:

N = 101

Output:

2

Explanation:

$1 + 0 + 1 = 2$, which is already a single digit.

18. Find the Sum of Squares of Digits

Problem Statement

Given an integer N, calculate the sum of the squares of all its digits.

Input

A single integer N.

Output

Print the sum.

Examples

Input:

N = 123

Output:

14

Explanation:

$$1^2 + 2^2 + 3^2 = 1 + 4 + 9 = 14.$$

19. Check Whether a Number is Prime

Problem Statement

Given an integer N, determine whether it is a prime number. A prime number is greater than 1 and has exactly two factors: 1 and itself.

Input

A single integer N.

Output

Print Prime or Not Prime.

Examples

Input:

N = 17

Output:

Prime

Explanation:

17 has no divisor other than 1 and 17.

Input:

N = 20

Output:

Not Prime

Explanation:

20 has factors other than 1 and itself.

20. Print All Prime Numbers in a Given Range

Problem Statement

Given two integers L and R, print all prime numbers between them.

Input

Two integers L and R.

Output

Print all prime numbers in the range [L, R].

Examples

Input:

L = 10, R = 30

Output:

11 13 17 19 23 29

Explanation:

These are the prime numbers between 10 and 30.

21. Find the Prime Factors of a Number

Problem Statement

Given an integer N, print all its prime factors.

Input

A single integer N.

Output

Print the prime factors, including repeated factors.

Examples

Input:

N = 60

Output:

2 2 3 5

Explanation:

$60 = 2 \times 2 \times 3 \times 5$.

22. Express a Number as the Sum of Two Prime Numbers

Problem Statement

Given an integer N, determine whether it can be expressed as the sum of two prime numbers.

Input

A single integer N.

Output

Print True if such a pair exists; otherwise print False.

Examples

Input:

N = 74

Output:

True

Explanation:

$74 = 3 + 71$, and both 3 and 71 are prime.

Input:

N = 11

Output:

False

Explanation:

11 cannot be expressed as the sum of two prime numbers.

23. Check Whether a Number is an Armstrong Number

Problem Statement

Given an integer N, determine whether it is an Armstrong number. For an N-digit number, calculate the sum of each digit raised to the power of the number of digits. If the sum equals the original number, it is an Armstrong number.

Input

A single integer N.

Output

Print Armstrong Number or Not Armstrong Number.

Examples

Input:

N = 153

Output:

Armstrong Number

Explanation:

$$1^3 + 5^3 + 3^3 = 1 + 125 + 27 = 153.$$

24. Check Whether a Number is a Perfect Number

Problem Statement

A number is called a Perfect Number if the sum of its proper divisors is equal to the number itself.

Input

A single integer N.

Output

Print Perfect Number or Not Perfect Number.

Examples

Input:

N = 28

Output:

Perfect Number

Explanation:

Proper divisors are 1, 2, 4, 7, and 14. Their sum is 28.

25. Check Whether a Number is an Abundant Number

Problem Statement

A number is called an Abundant Number if the sum of its proper divisors is greater than the number itself.

Input

A single integer N.

Output

Print Abundant Number or Not Abundant Number.

Examples

Input:

N = 18

Output:

Abundant Number

Explanation:

Proper divisors are 1, 2, 3, 6, and 9. Their sum is 21, which is greater than 18.

26. Check Whether a Number is Perfect, Abundant, or Deficient

Problem Statement

Given an integer N , classify it based on the sum of its proper divisors. $\text{Sum} = N \rightarrow \text{Perfect}$, $\text{Sum} > N \rightarrow \text{Abundant}$, $\text{Sum} < N \rightarrow \text{Deficient}$.

Input

A single integer N .

Output

Print Perfect, Abundant, or Deficient.

Examples

Input:

$N = 28$

Output:

Perfect

Explanation:

The proper-divisor sum is 28.

Input:

$N = 18$

Output:

Abundant

Explanation:

The proper-divisor sum is 21, greater than 18.

Input:

$N = 10$

Output:

Deficient

Explanation:

The proper-divisor sum is 8, smaller than 10.

27. Check Whether a Number is a Strong Number

Problem Statement

A Strong Number is a number in which the sum of the factorials of its digits is equal to the original number.

Input

A single integer N.

Output

Print Strong Number or Not Strong Number.

Examples

Input:

N = 145

Output:

Strong Number

Explanation:

$$1! + 4! + 5! = 1 + 24 + 120 = 145.$$

28. Check Whether a Number is a Harshad Number

Problem Statement

A number is called a Harshad Number (or Niven Number) if it is divisible by the sum of its digits.

Input

A single integer N.

Output

Print Harshad Number or Not Harshad Number.

Examples

Input:

N = 378

Output:

Harshad Number

Explanation:

Digit sum = $3 + 7 + 8 = 18$, and $378 \% 18 = 0$.

29. Check Whether a Number is an Automorphic Number

Problem Statement

A number is called Automorphic if its square ends with the number itself.

Input

A single integer N.

Output

Print Automorphic Number or Not Automorphic Number.

Examples

Input:

N = 25

Output:

Automorphic Number

Explanation:

$25^2 = 625$, whose last two digits are 25.

Input:

N = 76

Output:

Automorphic Number

Explanation:

$76^2 = 5776$, whose last two digits are 76.

30. Find the Factorial of a Number

Problem Statement

Given a non-negative integer N , find its factorial. Factorial is defined as $N! = N \times (N-1) \times \dots \times 2 \times 1$.

Input

A non-negative integer N .

Output

Print $N!$.

Examples

Input:

$N = 5$

Output:

120

Explanation:

$5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$.

Input:

$N = 0$

Output:

1

Explanation:

By definition, $0! = 1$.

31. Calculate the Power of a Number

Problem Statement

Given two integers X and N, calculate X raised to the power N.

Input

Two integers X and N.

Output

Print X^N .

Examples

Input:

$X = 2, N = 5$

Output:

32

Explanation:

$2 \times 2 \times 2 \times 2 \times 2 = 32.$

32. Find the Sum of an Arithmetic Progression

Problem Statement

Given the first term a , common difference d , and number of terms n , find the sum of the AP.

Input

Three integers: a , d , and n .

Output

Print the sum of the AP.

Examples

Input:

$a = 2, d = 5, n = 8$

Output:

156

Explanation:

The series is 2, 7, 12, 17, 22, 27, 32, 37 and its sum is 156.

33. Find the Sum of a Geometric Progression

Problem Statement

Given the first term a , common ratio r , and number of terms n , find the sum of the GP.

Input

Three integers: a , r , and n .

Output

Print the sum.

Examples

Input:

$a = 2, r = 3, n = 4$

Output:

80

Explanation:

$2 + 6 + 18 + 54 = 80$.

34. Print the Fibonacci Series

Problem Statement

Print the first N Fibonacci numbers. The sequence starts with 0, 1, 1, 2, 3, 5, 8, ... Each term is the sum of the previous two terms.

Input

An integer N.

Output

Print the first N terms.

Examples

Input:

N = 7

Output:

0 1 1 2 3 5 8

Explanation:

Each term after the first two is obtained by adding the previous two terms.

35. Find the GCD of Two Numbers

Problem Statement

Given two integers A and B, find their Greatest Common Divisor (GCD).

Input

Two integers A and B.

Output

Print the GCD.

Examples

Input:

A = 48, B = 18

Output:

6

Explanation:

The greatest common divisor of 48 and 18 is 6.

36. Find the LCM of Two Numbers

Problem Statement

Given two integers A and B, find their Least Common Multiple (LCM).

Input

Two integers A and B.

Output

Print the LCM.

Examples

Input:

A = 12, B = 18

Output:

36

Explanation:

36 is the smallest positive number divisible by both 12 and 18.

37. Count the Number of Factors of a Number

Problem Statement

Given an integer N, find the total number of positive factors of N.

Input

A single integer N.

Output

Print the number of factors.

Examples

Input:

N = 12

Output:

6

Explanation:

The factors of 12 are 1, 2, 3, 4, 6, and 12.

38. Find the Sum of All Factors of a Number

Problem Statement

Given an integer N, find the sum of all its positive factors.

Input

A single integer N.

Output

Print the sum of its factors.

Examples

Input:

N = 12

Output:

28

Explanation:

$1 + 2 + 3 + 4 + 6 + 12 = 28.$

39. Check Whether Two Numbers are Co-Prime

Problem Statement

Two numbers are called co-prime if their GCD is 1. Given two integers, determine whether they are co-prime.

Input

Two integers A and B.

Output

Print Co-Prime or Not Co-Prime.

Examples

Input:

A = 8, B = 15

Output:

Co-Prime

Explanation:

$\text{GCD}(8, 15) = 1$, so the numbers are co-prime.

40. Find the GCD of an Array of Numbers

Problem Statement

Given an array of integers, find the GCD of all its elements.

Input

An integer N followed by N integers.

Output

Print the GCD of all elements.

Examples

Input:

Array = [24, 36, 60]

Output:

12

Explanation:

$\text{GCD}(24, 36, 60) = 12$.

41. Check Whether a Number is a Neon Number

Problem Statement

A number is called a Neon Number if the sum of the digits of its square is equal to the number itself.

Input

A single integer N.

Output

Print Neon Number or Not Neon Number.

Examples

Input:

N = 9

Output:

Neon Number

Explanation:

$9^2 = 81$ and $8 + 1 = 9$.

42. Check Whether a Number is a Spy Number

Problem Statement

A number is called a Spy Number if the sum of its digits is equal to the product of its digits.

Input

A single integer N.

Output

Print Spy Number or Not Spy Number.

Examples

Input:

N = 1124

Output:

Spy Number

Explanation:

Sum = $1 + 1 + 2 + 4 = 8$ and product = $1 \times 1 \times 2 \times 4 = 8$.

43. Check Whether a Number is a Duck Number

Problem Statement

A Duck Number is a number that contains at least one zero in its digits.

Input

A positive integer N.

Output

Print Duck Number or Not Duck Number.

Examples

Input:

N = 1023

Output:

Duck Number

Explanation:

The number contains the digit 0.

44. Check Whether a Number is a Disarium Number

Problem Statement

A number is called a Disarium Number if the sum of its digits raised to their respective positions equals the number. Positions start from 1 from the left.

Input

A single integer N.

Output

Print Disarium Number or Not Disarium Number.

Examples

Input:

N = 135

Output:

Disarium Number

Explanation:

$$1^1 + 3^2 + 5^3 = 1 + 9 + 125 = 135.$$

45. Check Whether a Number is a Happy Number

Problem Statement

A number is called a Happy Number if repeatedly replacing the number by the sum of the squares of its digits eventually produces 1. If the process enters a cycle without reaching 1, it is not happy.

Input

A single integer N.

Output

Print Happy Number or Not Happy Number.

Examples

Input:

N = 19

Output:

Happy Number

Explanation:

$19 \rightarrow 82 \rightarrow 68 \rightarrow 100 \rightarrow 1$, so 19 is a Happy Number.

46. Calculate the Area of a Circle

Problem Statement

Given the radius of a circle, calculate its area.

Input

A positive number R representing the radius.

Output

Print the area of the circle.

Examples

Input:

R = 5

Output:

78.54

Explanation:

Area = $\pi \times R^2 = \pi \times 25 \approx 78.54$.

47. Find the Roots of a Quadratic Equation

Problem Statement

Given the coefficients a , b , and c of $ax^2 + bx + c = 0$, find its roots. Here a must not be zero.

Input

Three real numbers: a , b , and c .

Output

Print the roots and classify them as Real and Different, Real and Equal, or Complex.

Examples

Input:

$a = 1, b = -3, c = -10$

Output:

Real and Different; Roots = 5, -2

Explanation:

Discriminant = 49, so the roots are real and different.

Input:

$a = 1, b = 1, c = 1$

Output:

Complex

Explanation:

Discriminant = -3, so the roots are complex.

48. Calculate Permutations: N People Occupying R Seats

Problem Statement

Given N people and R seats, find the number of possible arrangements when order matters.

Input

Two integers N and R.

Output

Print NPR.

Examples

Input:

N = 5, R = 3

Output:

60

Explanation:

$5P3 = 5! / (5-3)! = 60$.

49. Calculate Combinations: Select R People from N People

Problem Statement

Given N people and R selections, find the number of ways to select R people from N people when order does not matter.

Input

Two integers N and R.

Output

Print NCR.

Examples

Input:

N = 5, R = 2

Output:

10

Explanation:

$5C2 = 5! / (2! \times 3!) = 10.$

50. Add Two Fractions and Simplify the Result

Problem Statement

Given the numerator and denominator of two fractions, calculate their sum and return the result in its simplest form.

Input

Four integers: Numerator1, Denominator1, Numerator2, and Denominator2.

Output

Print the numerator and denominator of the simplified result.

Examples

Input:

Numerator1 = 3, Denominator1 = 4, Numerator2 = 1, Denominator2 = 7

Output:

Numerator = 25, Denominator = 28

Explanation:

$$3/4 + 1/7 = (3 \times 7 + 1 \times 4) / (4 \times 7) = 25/28.$$

Input:

Numerator1 = 5, Denominator1 = 2, Numerator2 = 1, Denominator2 = 2

Output:

Numerator = 3, Denominator = 1

Explanation:

$$5/2 + 1/2 = 6/2 = 3/1.$$